

Alex J. Chan

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I am interested in developing **safe** machine learning and AI systems that work with humans effectively. My work has often explored **(inverse) reinforcement learning** and **imitation learning** in order to learn from humans, and I'm interested in using this knowledge when applied to **large language/multimodal models** to develop **personalised, human-centric**, decision systems.

EXPERIENCE

Convergence Labs Ltd.

London, UK

Founding Member of Technical Staff (Research Scientist)

Sep. 2024 – Present

- As one of the earliest employees, I'm building our capacity to train and serve our own vision-language(-action) models on our large scale multi-node H100 cluster, as well as guiding our research agenda.
- Developing RL techniques for improving agentic behaviour in models for web-browsing and automated workflows.

Spotify

London, UK

Research Scientist - Tech Research (Personalization Mission)

Mar. - Jun. 2024

- Working on reinforcement learning training of LLMs on domains including tool use and controllable generation.
- Building scalable RL pipelines for internal LLMs using Ray and vLLM for distributed training and serving.

Research Scientist Intern

Jun. - Oct. 2023

- Created various cultural media datasets through web-scraping in order to finetune a family of models to be more aware of transient news and location/cultural specific issues to improve detection of potential violative content.
- Incorporated these cultural models as tools for larger models to generate personalised guidance to content moderators, especially in the case when there is a cultural mismatch between the moderator and original content.
- Project resulted in writing an academic paper, filing for a patent, and I received a full-time return offer.

Stanford Existential Risk Initiative

Berkeley, CA

Researcher - Dr Owain Evans' Research Group

Nov. 2022 – Jun. 2023

- Investigated deceptive behaviour in LLMs and possible detection with only black-box query knowledge.
- Fine-tuned and ran large (up to 65 billion parameters) models on remote servers with extensive shell scripting using various distributed computing packages including PyTorch Distributed and DeepSpeed.

Microsoft Research

Cambridge, UK

PhD Scholar

Oct. 2020 – Present

- Student researcher co-supervised by Dr Aditya Nori (previously Dr Danielle Belgrave before she joined DeepMind) “A Smart Care System for Healthcare using Contextual Reinforcement Learning”.

University College London

London, UK

Research Assistant - Dr Sam Livingstone's Research Group

Jun. – Aug. 2018

- Implemented a variety of MCMC samplers and practical convergence diagnostic methods in Python, running experiments to compare their output with recently proposed theoretical bounds.

EDUCATION

University of Cambridge

Cambridge, UK

PhD Machine Learning - Supervisor: Professor Mihaela van der Schaar

Oct. 2020 – Present

- Title: Aligning Models for Human-Centric Decision Systems.
- First-author publications in all three of the major machine learning conferences: ICML, NeurIPS, and ICLR.
- In total my published work has been cited more than 300 times, and I have an h-index of 9.

MPhil Machine Learning and Machine Intelligence

Oct. 2019 – Sep. 2020

- Awarded with Commendation and an average of 79%. 92% for my thesis “Interpretable Policy Learning” - developing interpretable imitation learning algorithms for decision making in high stakes environments.
- Taught component focusing on probabilistic machine learning, with modules on natural language processing (NLP), reinforcement learning (RL), and computational neuroscience.

University College London

London, UK

BSc Statistics

Oct. 2016 – June 2019

- 1st Class Honours - 81% average. Achieved 84% on my final year project on probabilistic deep learning, focusing on flexible Bayesian approximations in deep neural networks.
- Modules include significant mathematical and statistical content covering probability, linear models, mathematical analysis, and advanced linear algebra.

CONFERENCE PUBLICATIONS

Discovering Preference Optimization Algorithms with and for Large Language Models

C. Lu, S. Holt, C. Fanconi, **A. J. Chan**, J. Foerster, M. van der Schaar, and R. T. Lange. *Advances in Neural Information Processing Systems (NeurIPS)* 2024.

Dense Reward for Free in Reinforcement Learning from Human Feedback

A. J. Chan, H. Sun, S. Holt, and M. van der Schaar. *International Conference on Machine Learning (ICML)* 2024.

How to Catch an AI Liar: Lie Detection in Black-box LLMs by Asking Unrelated Questions

L. Pacchiardi*, **A. J. Chan***, S. Mindermann, I. Moscovitz, A. Pan, Y. Gal, O. Evans, and J. M. Brauner. *International Conference on Learning Representations (ICLR)* 2024.

AllSim: Systematic Simulation and Benchmarking of Repeated Resource Allocation Policies in Multi-User Systems with Varying Resources

J. Berrevoets, D. Jarrett, **A. J. Chan**, and M. van der Schaar. *Proceedings of the Neural Information Processing Systems (NeurIPS) track on Datasets and Benchmarks* 2023.

GAUCHE: A Library for Gaussian Processes in Chemistry

R. Griffiths, L. Klarner, H. Moss, A. Ravuri, S. T. Truong, Y. Du, S. Don Stanton, G. Tom, B. Ranković, A. R. Jamasb, A. Deshwal, J. Schwartz, A. Tripp, G. Kell, S. Frieder, A. Bourached, **A. J. Chan**, J. Moss, C. Guo, J. P. Dürholt, S. Chaurasia, J. W. Park, F. Strieth-Kalthoff, A. Lee, B. Cheng, A. Aspuru-Guzik, P. Schwaller, J. Tang. *Advances in Neural Information Processing Systems (NeurIPS)* 2023.

Synthetic Model Combination: An Instance-wise Approach to Unsupervised Ensemble Learning

A. J. Chan and M. van der Schaar. *Advances in Neural Information Processing Systems (NeurIPS)* 2022.

Inverse Online Learning: Understanding Non-Stationary and Reactionary Policies

A. J. Chan, A. Curth, and M. van der Schaar. *International Conference on Learning Representations (ICLR)* 2022.

POETREE: Interpretable Policy Learning with Adaptive Decision Trees

A. Pace, **A. J. Chan**, and M. van der Schaar. *International Conference on Learning Representations (ICLR)* 2022.

The Medkit-learn(ing) Environment: Medical Decision Modelling through Simulation

A. J. Chan, I. Bica, A. Hüyük, D. Jarrett, and M. van der Schaar. *Proceedings of the Neural Information Processing Systems (NeurIPS) track on Datasets and Benchmarks* 2021.

Scalable Bayesian Inverse Reinforcement Learning

A. J. Chan and M. van der Schaar. *International Conference on Learning Representations (ICLR)* 2021.

Generative Time Series Modelling with Fourier Flows

A. M. Alaa, **A. J. Chan**, and M. van der Schaar. *International Conference on Learning Representations (ICLR)* 2021.

Unlabelled Data Improves Bayesian Uncertainty Calibration under Covariate Shift

A. J. Chan, A. M. Alaa, Z. Qian, and M. van der Schaar. *International Conference on Machine Learning (ICML)* 2020.

JOURNAL PUBLICATIONS

When is Off-Policy Evaluation Useful in Contextual Bandits? A Data-Centric Perspective

Hao Sun*, **A. J. Chan***, Nabeel Seedat, A. Hüyük, and M. van der Schaar. *Journal of Data-centric Machine Learning Research (DMLR)* 2024.

Synthetic Model Combination: A New Machine Learning Method for Pharmacometric Model Ensembling

A. J. Chan, R. Peck, M. Gibbs, and M. van der Schaar. *CPT: Pharmacometrics & Systems Pharmacology* 2023.

WORKSHOP PUBLICATIONS

Actions Speak Louder than Words: Superficial Fairness Alignment in LLMs

Q. Wei, **A. J. Chan**, L. Goetz, D. Watson, and M. van der Schaar. *ICLR Workshop on Reliable and Responsible Foundation Models* 2024.

Optimising Human-AI Collaboration by Finding Convincing Explanations

A. J. Chan, A. Hüyük, and M. van der Schaar. *NeurIPS XAI in Action* 2023.

*Equal contribution.

Practical Approaches for Fair Learning with Multitype and Multivariate Sensitive Attributes

T. Liu, **A. J. Chan**, B. van Breugel, and M. van der Schaar. *NeurIPS Algorithmic Fairness through the Lens of Causality and Privacy (AFCP)* 2022.

PREPRINTS / UNDER-REVIEW

Harmonizing Global Voices: Culturally-Aware Models for Enhanced Content Moderation

A. J. Chan, J. L. R. García, F. Silvestri, C. O'Donnel, and K. Palla. <https://arxiv.org/abs/2312.02401>.

AWARDS/PRIZES

OpenAI Superalignment Fellowship (150k USD)

- Grant for research project on activation steering generalisation in large language models.

Machine Learning Alignment Theory Scholarship (20k USD)

- Scholarship funding for research project on how large language models can lie when articulating decision rules.

Microsoft Research PhD Scholarship (≈160k GBP)

- Received the award for full funding of my PhD co-supervised with Microsoft Research (Dr Aditya Nori, and Dr Danielle Belgrave) “A Smart Care System for Healthcare using Contextual Reinforcement Learning”.

G-Research PhD Prize in Maths and Data Science (7k GBP)

- Runner up in the G-Research competition for best draft PhD dissertation.

EPSRC Vacation Grant (≈2k GBP)

- Awarded funding grant by the Engineering and Physical Sciences Research Council to conduct a research project during the summer on Markov chain Monte Carlo methods.

SKILLS

Machine Languages: Python, R, MATLAB, PostgreSQL, HTML.

Human Languages: English, Conversational French.

Libraries/Tools: PyTorch, JAX, TensorFlow, Transformers, DeepSpeed, pandas, NumPy, TRL, Azure, GCP, Ray.

SUPERVISION

University of Cambridge Cambridge, UK

MPhil Machine Learning and Machine Intelligence Theses Mar. – Aug. 2021

- **Tennison Liu:** *Fair Policy Learning*. (Work published at AFCP 2022).
- **Alizée Pace:** *Adaptive Decision Tree Policies* (Resulted in a Spotlight at ICLR 2022).

University of Oxford Oxford, UK

MSc Statistical Science Thesis Mar. – Aug. 2021

- **Yuling Chen:** *Clustered Bayesian Inverse Reinforcement Learning Via Variational Inference*.

SERVICE

AAAI Workshop on Representation Learning for Responsible Human-Centric AI

Invited Area Chair 2023

NeurIPS SyntheticData4ML Workshop

Program Committee / Area Chair 2022

NeurIPS Workshop on Causality for Real-world Impact

Invited Reviewer 2022

ICML/ICLR/NeurIPS

Invited Reviewer ICML21-23, NeurIPS21-23, ICLR21-24 2021 – Present

Code First Girls

Volunteer course instructor for “Introduction to Python Programming” Jan. – March 2021

University of Cambridge

Club Captain, Wolfson College Boat Club Aug. 2021 – Aug. 2022

- As Captain of the boat club, I was in charge of the overall running of the club, organising the training of the members as well as broader events and the alumni network.

University College London

London, UK

Vice President/Treasurer - Pure Krav Maga Society

Oct. 2018 – June 2019

- I oversaw the organisation and finances behind sessions while helping to run classes as a trainee instructor.

Welfare Officer, Effective Altruism Society

Oct. 2018 – June 2019

- I was responsible for engaging with the wider community to develop more of an understanding of the aims of Effective Altruism as well as looking out for the welfare of our members and helping develop the society further.

Electric Eels Swimming Club

Windsor, UK

Volunteer Swimming Coach

2011 – 2015

- I spent four years volunteering with the club, which aims to provide special coaching for children with Down syndrome, coaching both groups and 1-on-1 at a range of swimming ability
- I became ASA certified in Teaching Aquatics, allowing me to develop my technical and communication skills to be a more effective coach.